

PROGRAMME : Diploma Programme in IF / CM
COURSE : Python Programming

COURSECODE : 232405

LEARNING AND ASSESSMENT SCHEME:

Learning Scheme						Credits	Assessment Scheme											Total Marks
Actual Contact Hrs./Week			SLH	NLH	Paper Duration		Theory				Based on LL and TSL				Based on self Learning			
											Practical							
CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA			
03	01	02	00	06	03	03	Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	150	
							30	70	100	40	25	10	25#	10	--	--		

IKS Content: Total Learning Hours for Term: 0 Hrs

Indicates Online Examination, @ Internal assessment

1.0 RATIONALE:

Python is an open source, general-purpose and most versatile programming language. Python code is simple, readable, short, intuitive, and powerful, and thus it is effective for introducing computing and problem solving for beginners. This course covers basic fundamentals of Python programming, which also provides a foundation for further exploration of its more advanced applications in a variety of domains, including application development, data science, artificial intelligence, machine learning, and more.

2.0 INDUSTRY / EMPLOYER EXPECTED OUTCOME:

The Aim of this course is to help the students to attain these industry-identified outcomes through various teaching learning experiences:

Develop applications using python to solve given problem.

3.0 COURSE OUTCOMES:

The course content should be taught and learning imparted in such a manner that students are able to acquire required learning outcome in cognitive, psychomotor and affective domain to demonstrate following course outcomes:

1. Develop python programs using control flow statements.
2. Perform operations on various data structures in Python.
3. Develop packages to solve given problem using python.
4. Apply object-oriented approach to solve given problem using python.
5. Use relevant built-in python package to develop application.

4.0 COURSE DETAILS:

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
Unit-I Introduction to Python and Control flow statements	1a. Explain given feature of python. 1b. Write python program to perform basic input output operations. 1c. Write python program to solve given expression. 1d. Implement given decision making	1.1 Introduction: Features, History and Applications of Python, Python IDE's 1.2 Python building blocks: Indentation, Identifiers, Variable, Comments, Keywords 1.3 Basic input output operations: input(), print()	06

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
	statements and looping statements in python program.	1.4 Operators: Arithmetic, Relational, Assignment, Logical, Bitwise, Membership and Identity operator 1.5 Control flow statements: Conditional statements (if, if-else, if-elif-else, nested if), Loops in python (while, for, nested loops), Loop manipulation 1.6 statements (continue, pass, break, else)	
Unit-II Data Structures in Python	2a. Write python program to manipulate lists. 2b. Write python program to manipulate tuples. 2c. Write python program to manipulate sets. 2d. Write python program to manipulate dictionaries.	2.1 List: a) Defining lists, accessing values from list, deleting list values, updating lists b) Basic list operations c) Built-in list functions/methods 2.2 Tuple: a) Defining Tuple, accessing values from Tuple b) Basic Tuple operations c) Built in Tuple functions/methods 2.3 Set: a) Defining Sets, accessing values from set, deleting set values b) Basic set operations c) Built in set functions/methods 2.4 Dictionary: a) Defining Dictionary, accessing values from Dictionary, deleting Dictionary values, updating Dictionary b) Basic Dictionary operations Built in Dictionary functions/methods	08
Unit-III Functions, Modules and Packages in Python	3a. Write relevant user defined functions for the given problem. 3b. Write relevant user defined module for the given problem. 3c. Write packages for the given problem.	3.1 Functions: Defining function, Calling function, Function arguments, Return statement, Scope of Variable, Lambda functions 3.2 Modules: Create user defined Module, Importing a module, Using python built-in modules, Namespace and scoping 3.3 Python Packages: Create user defined Package, Importing a Package, Using python built-in Packages,	12

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
		Installing packages using PIP	
Unit-IV Object Oriented Programming in Python	4a. Write python program using classes and objects to solve given problem. 4b. Implement python program using different types of constructors. 4c. Write program to demonstrate polymorphism. 4d. Write python code using data abstraction for given problem. 4e. Apply inheritance for the given problem.	4.1 Object oriented Concepts: Creating class, Creating object 4.2 Constructors in python (Parameterized & Non-Parameterized), the self parameter 4.3 Polymorphism: Method Overloading and Overriding 4.4 Data Hiding / Abstraction Inheritance: Single Inheritance, Multiple Inheritance, Multilevel Inheritance	12
Unit-V Introduction to Built-in Packages in Python	5a. Write python program to use pandas package for the given problem. 5b. Create GUI application using tkinter package for the given problem. 5a. Create a python application to connect with database.	5.1 Pandas: Use of pandas, pandas series, pandas DataFrames, pandas Read CSV 5.2 Creating GUI using tkinter: Introduction to tkinter, Widgets (Entry, Label, Button, RadioButton, Checkbutton), Creating a simple GUI application 5.3 Connecting to Database using MySQL: Installing mysql-connector, cursor() object, execute() method, fetchall() method, Creating simple program to connect database	07
		TOTAL	45

5.0 SUGGESTED SPECIFICATION TABLE WITH MARKS (THEORY):

Sr.No	Unit	Unit Title	Aligned Cos	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Introduction to Python and Controlflow statements	CO1	12	4	6	6	18
2	II	Data Structures in Python	CO2	06	2	6	6	10
3	III	Functions, Modules and Packages in Python	CO3	08	2	4	6	12
4	IV	Object Oriented Programming in Python	CO4	08	4	4	4	12
5	V	Introduction to Built-in Packages in Python	CO5	11	4	6	6	18
Grand Total				45	16	26	28	70

6.0 LABORATORY LEARNING OUTCOME AND ALLIED PRACTICAL/ TUTORIAL EXPERIENCES:

The tutorial/practical/assignment/task should be properly designed and implemented with an attempt to develop different types of cognitive and practical skills (**Outcomes in cognitive, psychomotor and affective domain**) so that students are able to acquire the desired programme outcome/course outcome.

Note: Here only outcomes in psychomotor domain are listed as practical/exercises. However, if these practical/exercises are completed appropriately, they would also lead to development of **Programme Outcomes/Course Outcomes in Affective domain** as given in the mapping matrix for this course. Faculty should ensure that students also acquire Programme Outcomes/Course Outcomes related to affective domain.

Sr. No.	Unit No.	LLO	Practical Exercises (Outcomes in Psychomotor Domain)	Hours
01	I	1.a Install the given Python IDE.	Install given Python IDE.	02
02	II	2.a Use different data types in a Python program.	Write and execute program using different data types.	02
03	II	3.a Write Python program to solve given expression.	Implement a python programs using following operators: 1. Arithmetic 2. Logical 3. Bitwise	04
04	III	4.a Write python program for solving given problem using various if statements	Implement a python program to demonstrate use of conditional statements (if, if.....else) and Loop control statements(while loop or for loop).	02

Sr. No.	Unit No.	LLO	Practical Exercises (Outcomes in Psychomotor Domain)	Hours
		4.b Write python program for solving given problems using appropriate loops.		
05	III	6.a Write python program to perform operations on list.	Implement a Python program to perform following operations on list: a)Create list b) access list c)update list d)delete list.	04
06	III	7.a Write python program to perform operations on tuple.	Write a program to perform following operations on a tuple a)Create b)Access c)Update d)delete	04
07	IV	8.a Write function to solve given problem.	Write a program to print factors of given number using function.	02
08	IV	9.a Write a program for given problem using classes and objects.	Program to add and subtract two complex numbers using class and objects	02
09	IV	10.a Use panda package and its appropriate functions/methods to solve a given problem.	Implement Python program to perform following operations using panda package: 1. Create Series from Array 2. Create Series from List 3. Access element of series 4. Create DataFrame using List or dictionary	04
10	V	14.a Write python program to connect database and retrieve data.	Write program to create a connection between database and python, select records from database table and display the result	04
				30

7.0 SELF LEARNING: NA

8.0 SPECIAL INSTRUCTIONAL STRATEGIES (If any):

These are sample strategies, which the teacher can use to accelerate the attainment of the various outcomes in this course

1. Use online tools to improve their learning, such as the e-Kumbh from AICTE and the Virtual Labs from IIT.
2. Encourage students to participate in various coding competitions, such as hackathons, online coding contests on websites like hackerrank, Codechef etc.

9.0 ASSESSMENT METHODOLOGY

Formative Assessment-PR
Lab Performance, Viva/voce

Summative Assessment-PR
Lab Performance, Viva/voce

10.0 LEARNING RESOURCES:

A) Books:

Sr.No.	Author	Title of Book	Publication
1	R. Nageswara Rao	Core Python Programming	Dreamtech Press, ISBN-13:9789390457151
2	Mark Lutz	Learning Python	O'Reilly Media, Inc, ISBN: 9781449355739
3	David Amos, Dan Bader, Joanna Jablonski, Fletcher Heisler	Python Basics	Real Python, ISBN-13: 9781775093329
4	Dr. Jeeva Jose	Taming Python by Programming	Khanna Book Publishing CO(P) LTD, New Delhi, ISBN: 9789386173348
5	Rupesh Nasre	Python Programming	AICTE, ISBN 9788195986354 [Online

B) Software/Learning Websites:

- 1) <https://ekumbh.aicte-india.org/allbook.php>
- 2) <https://python-iitk.vlabs.ac.in/>
- 3) <https://spoken-tutorial.org/watch/Python+3.4.3/Input-output/English/>
- 4) https://onlinecourses.nptel.ac.in/noc19_cs41/preview
- 5) https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0130944397935001602592_shared/overview
- 6) <https://www.geeksforgeeks.org/python-gui-tkinter/>
- 7) https://www.w3schools.com/python/python_mysql_getstarted.asp
- 8) https://www.tutorialspoint.com/python_pandas/index.htm
- 9) <https://www.programiz.com/python-programming/object-oriented-programming>

C) Major Equipment/ Instrument with Broad Specifications:

- 1) Any Database Software
- 2) Computer System-(Any computer system with basic configuration)
- 3) Python Interpreter / IDE-(Any open source python distribution such as anaconda etc)
(Any open source IDE such IDLE, Jupyter Notebook, Spyder, PyCharm etc)

11.0 MAPPING MATRIX OF PO's, CO's and PSO's:

Course Outcomes	Programme Outcomes (PO's)							Programme Specific Outcomes (PSO's)			
	1	2	3	4	5	6	7	1	2	3	4
CO1	M	L	L	L	-	-	-	H	L	M	M
CO2	M	L	L	L	-	-	-	M	-	M	-
CO3	H	M	M	M	-	-	-	M	M	H	-
CO4	H	H	H	M	-	-	L	M	H	-	H
CO5	H	M	H	H	-	-	L	M	-	H	-

H: High Relationship, M: Moderate Relationship, L: Low Relationship.

6.0 SUGGESTED QUESTION PAPER PROFILE:

UNIT No	CO	Marks Per Unit	1.35 Times Marks	Question Number Wise Marks						Actual Distribution of Marks
				01	02	03	04	05	06	
I	CO1	18	24	4	4	4		8	4	24
II	CO2	10	14	2	4	4	4			14

III	C03	12	16	4	4	4	4			16
IV	C04	12	16	4	4	4	4			16
V	C05	18	24				4	8	12	24
	Total	70	94	14	16	16	16	16	16	94

